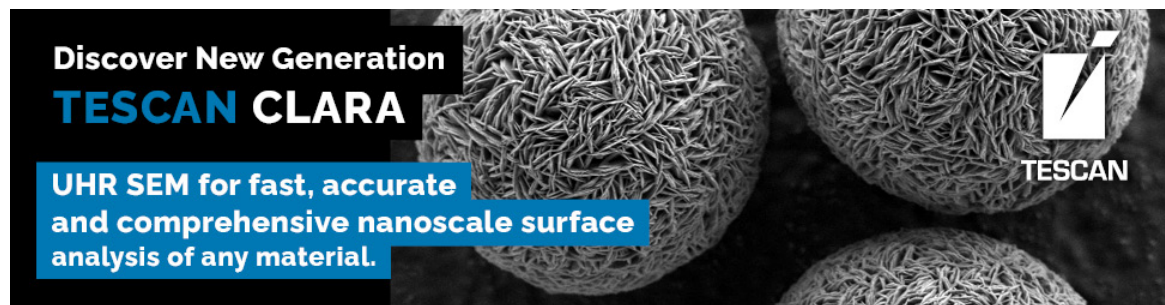


Characterization of Graphene Coatings on 8620 Alloy Additive Manufactured Steel

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Proceedings

Characterization of Graphene Coatings on 8620 Alloy Additive Manufactured Steel

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Selective laser melting (SLM) is one of the additive manufacturing (AM) processes which create dense components with high control over the alloy compositions [1]. The 8620 alloy is a low carbon, nickel chromium molybdenum alloy steel which is typically used for machine parts. However, its high iron content (~98%) makes it very susceptible to oxidation. This research focuses on developing graphene surface coatings for 8620 and other iron-rich steels. In this study, AM8620 samples made by SLM were used. Previously, we demonstrated the coating of steel surfaces by a liquid-exfoliated graphene flakes (EGFs) method [2]. In this report, we present a graphene coating formed by directly growing graphene on the AM steel surface using a chemical vapor deposition (CVD) method. The effects of the surface morphologies and microstructure of the AM8620 steel on graphene coatings in relation to the two coating methods were characterized.

SLM samples of (1cm)³ were polished with 120, 600, 1200 grit sandpapers sequentially, followed by polycrystalline diamond particles of 6µm then 0.25µm, and last by 0.04µm colloidal silica. A 3% nitric acid in ethanol solution was used to etch the surface. A digital Keyence VHX-7000 microscope, an SEM (SNE-4500M-Plus), and a Horiba Jobin Yvon HR800 Raman spectrometer were used for characterization.

Fig. 1 shows the microstructure of AM8620 samples and CVD graphene. In fig. 1(a), the optical image shows the characteristic pattern of melt pool boundaries (MPBs) from the laser. The enlarged area from fig. 1(a) is shown in fig. 1(b), where the martensitic structure of AM steel is apparent. The SEM image, fig. 1(c), shows the ferrite and cementite exposed by selective etching. Fig. 1(d) is an SEM image of a polished and etched AM8620 sample coated with graphene by direct CVD growth. Note that the microstructure changes between (c) and (d) were caused by the CVD process, in which samples were heated to 650°C and then exposed to hydrocarbon gas to catalyze the CVD reaction. A thin layer of graphene coating over the newly formed grains is visible and the surface coverage is optimum.

Fig. 2 compares the CVD graphene coating with EGF coatings. Fig. 2(a) is the AM8620 surface which has been polished to improve the EGF adhesion. The EGF coating in fig. 2(b) is composed of large multilayer sheets of graphene; with individual pieces covering large areas of the substrate. Though multilayer, the coating is still transparent to the electron beam and the coated surface can be seen through the EGFs. The Raman spectrum (blue curve) shown in fig. 2(c) is the averaged spectrum for CVD graphene, while the spectrum (red curve) is the averaged spectrum for the EGF coating. High-quality monolayer graphene has a ratio of the 2D and G peaks greater than one, as shown for the CVD graphene coating. The EGF coating shows a relatively lower 2D:G ratio, suggesting a multilayer coating. The Raman maps (obtained by taking 1600 spectra in a 40µm x 40µm area) shown in fig. 2(d) confirm the high coverage of both coatings (black pixels have signal to noise ratio less than 15 standard deviations). The 2D:G ratios of 0.15-0.6 confirmed the EGF coating as multilayer. The CVD graphene had some areas with a 2D:G ratio lower than 1, and most areas with ratios of 1-6 for monolayer graphene.

The multilayer EGF coatings have flexibility to conform with a rough AM8620 surface, and improved polishing of the surface allows better contact for EGF coatings. The CVD method enables monolayer graphene coatings. However, it is affected by material composition, grain size and structure change during catalyzed growth. Tuning the surface preparation and control of the sample microstructure will further increase the capabilities of EGF and CVD graphene coatings for 8620 steel and other alloys [3].

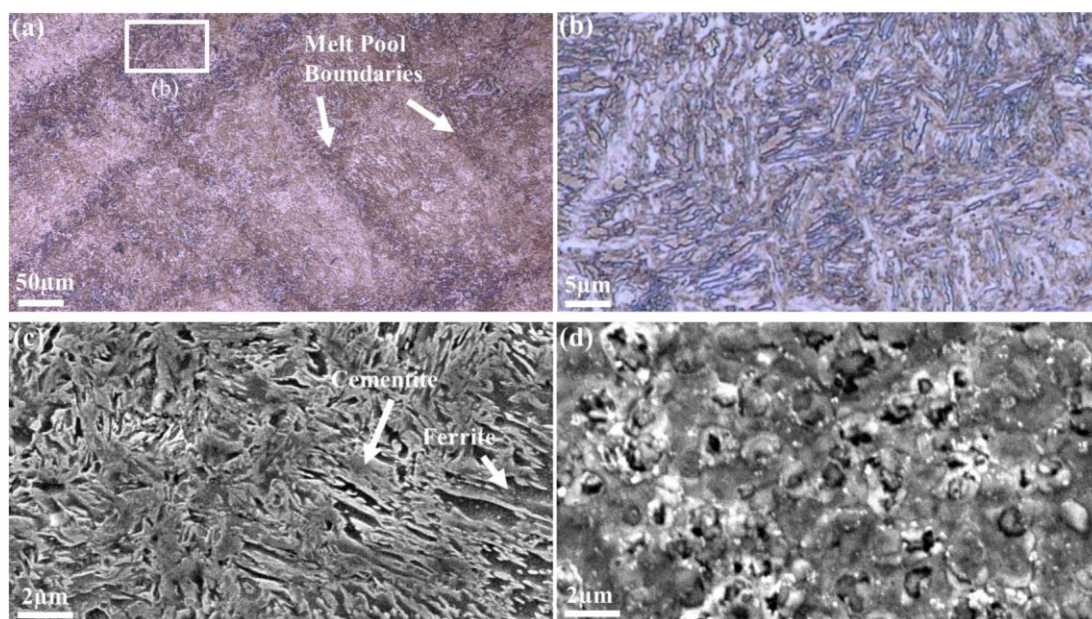


Fig. 1. Microstructure of polished and etched AM8620 steel. (a) Optical image of AM8620 with MPBs from the SLM process. (b) High resolution image from (a) with martensitic structure. (c) SEM image of AM 8620 structure before growth and (d) graphene grown directly on the AM8620 surface.

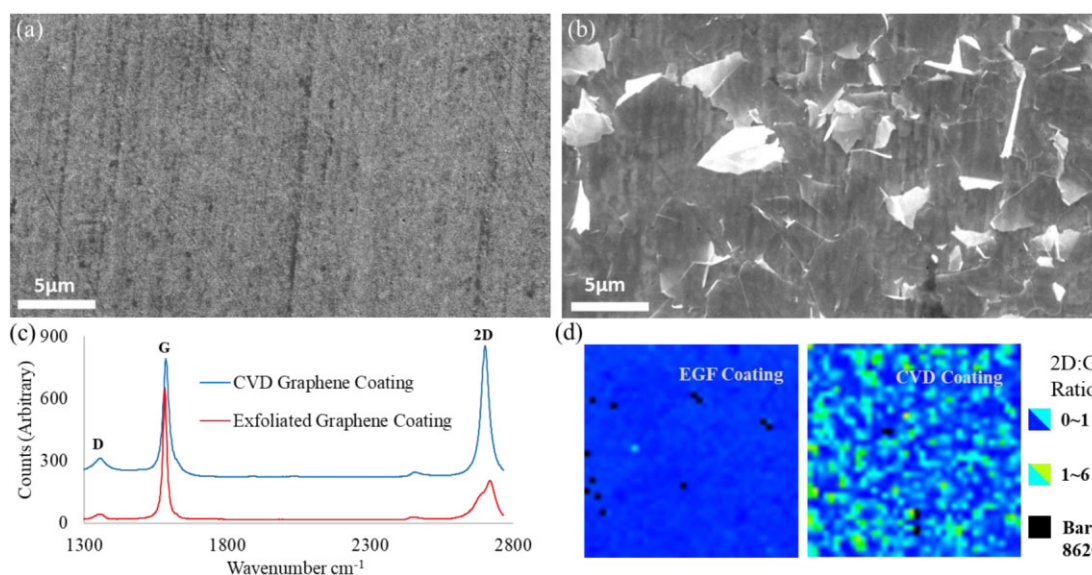


Fig. 2. EGF coating and comparison with CVD graphene coating. (a) Polished AM8620 before coating (b) SEM image of AM8620 coated with EGFs. (c) The Raman spectra (excited by a 532nm laser) of two graphene coatings. (d) Raman maps of 2D:G ratio for EGF coating and CVD graphene coating. Blue and teal areas have a ratio up to 1, while green and yellow represent a ratio greater than 1.

References

1. Schindelholz, et al., *Corrosion* 77(5) (2021), p. 484.
2. K Hood *et al.*, *Microscopy and Microanalysis* 28(S1) (2022), p. 920.
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